WEEK 1 ASSIGNMENT

Real Time Operating System

onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94

swaroopmandal@acpce.ac.in

NPTEL » Real Time Operating System

Announcements | About the Course | Q&A | Progress | Mentor | Review Assignment | Course Recommendations

Course outline

- About NPTEL
- How does an NPTEL online course work?
- Week 0
- Week 1
  - Lecture 1 : Introduction
  - Lecture 2 : Basics of Task scheduling
  - Lecture 3: Cyclic executives
  - Lecture 4: Cyclic Scheduler
  - Lecture 5 : Cyclic Scheduler (contd.)
  - Lecture material of Week 1
  - Quiz: Week 01: Assignment 01
  - Week 1 Feedback Form
- Week 2
- Week 3
- Week 4

Week 01: Assignment 01

The due date for submitting this assignment has passed.

Due on 2026-02-25, 23:59 IST.

Assignment submitted on 2026-02-25, 21:51 IST

1) The type of events produced by a smoke sensor in an industrial setting can be classified into which one of the following? *1 point*

a. Periodic
b. Aperiodic
c. Sporadic
d. Both aperiodic and sporadic
e. Neither aperiodic not sporadic

a.
b.
c.
d.
e.

Yes, the answer is correct.
Score: 1
Accepted Answers:
c.

2) Which one of the following is an important characteristic of a firm real-time task of an application? *1 point*

a. The result produced by the task is discarded if it is produced after its deadline
b. Application fails if the task produces a correct result, but after the deadline has

31°C Sunny | ENG IN | 88% | 10:19 AM 15/04/2026

Real Time Operating System

onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94

- Week 2
- Week 3
- Week 4
- Assignment Solutions
- Text Transcripts
- Books
- Live session - 2026

2) Which one of the following is an important characteristic of a firm real-time task of an application? *1 point*

a. The result produced by the task is discarded if it is produced after its deadline
b. Application fails if the task produces a correct result, but after the deadline has passed
c. The application fails if the task produces an incorrect result after the deadline has passed
d. Utility of the result produced by the task decreases with time, if the result is produced after the deadline
e. The earlier the result produced by a task, the higher its utility.

a.
b.
c.
d.
e.

Yes, the answer is correct.
Score: 1
Accepted Answers:
a.

3) Which of the following determines the order in which different tasks are executed in a cyclic scheduler? *1 point*

a. User inputs
b. Task deadlines
c. Predefined schedule table
d. Operating system randomness
e. Position in a dynamic task queue

a.
b.
c.
d.
e.

Yes, the answer is correct.
Score: 1
Accepted Answers:

Trending videos Margo & Edith i... | ENG IN | 89% | 10:20 AM 15/04/2026

Real Time Operating System -
Ask Gemini
onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94
Google Gemini ChatGPT DeepSeek - Into the... Generative AI for Be... Home | Adobe Course: Data Scienc... HackNiche 4.0 | Dev... All Bookmarks

4) Assume that a task is specified by the triplet (P,E,D), where P is the task period, E is the execution time, and D is the deadline. Consider three periodic real-time tasks are to be scheduled using EDF (Earliest Deadline First) on a uniprocessor: (100,30,100), (150,20,150), (200,25,200). What would be the processor utilization due to the three tasks? *1 point*

a. 0.5

b. 0.56

c. 0.6

d. 0.67

e. 0.7

- a.
- b.
- c.
- d.
- e.

Yes, the answer is correct.
Score: 1
Accepted Answers:
b.

5) A cyclic (time-driven) scheduler is used to schedule a set of periodic real-time tasks. Which one of the following statements is **TRUE** about cyclic schedulers? *1 point*

a. A cyclic scheduler dynamically selects the next task to execute based on task priorities at runtime.
b. In a cyclic scheduler, tasks may miss deadlines because task priorities are recomputed after every job release.
c. A cyclic scheduler executes tasks according to a predefined static schedule constructed off-line.
d. A cyclic scheduler guarantees optimal CPU utilization for any arbitrary set of periodic tasks.
e. A cyclic scheduler allows tasks to be preempted at any instant of time.

- a.
- b.

Trending videos
Margo & Edith i...
ENG IN 89% 10:20 AM 15/04/2026

Real Time Operating System -
Ask Gemini
onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94
Google Gemini ChatGPT DeepSeek - Into the... Generative AI for Be... Home | Adobe Course: Data Scienc... HackNiche 4.0 | Dev... All Bookmarks

5) A cyclic (time-driven) scheduler is used to schedule a set of periodic real-time tasks. Which one of the following statements is **TRUE** about cyclic schedulers? *1 point*

a. A cyclic scheduler dynamically selects the next task to execute based on task priorities at runtime.
b. In a cyclic scheduler, tasks may miss deadlines because task priorities are recomputed after every job release.
c. A cyclic scheduler executes tasks according to a predefined static schedule constructed off-line.
d. A cyclic scheduler guarantees optimal CPU utilization for any arbitrary set of periodic tasks.
e. A cyclic scheduler allows tasks to be preempted at any instant of time.

- a.
- b.
- c.
- d.
- e.

Yes, the answer is correct.
Score: 1
Accepted Answers:
c.

6) In a cyclic executive scheduler, a set of periodic tasks is scheduled using frames of equal length. Which of the following conditions are necessary for a frame size *f* to be feasible? *1 point*

a. *f* must be equal to the greatest common divisor (GCD) of all task periods
b. *f* must be greater than or equal to the largest task execution time
c. *f* must divide the hyperperiod of the task set
d. *f* must be less than or equal to the smallest task deadline
e. *f* must be an integer multiple of all task execution times

- a.
- b.
- c.
- d.
- e.

Yes, the answer is correct.
Score: 1
Accepted Answers:

Trending videos
Margo & Edith i...
ENG IN 89% 10:20 AM 15/04/2026

Real Time Operating System - ...
onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94
Ask Gemini
Google Gemini | ChatGPT | DeepSeek - Into the... | Generative AI for Be... | Home | Adobe | Course: Data Scienc... | HackNiche 4.0 | Dev... | All Bookmarks

7) In a cyclic executive scheduler, which of the following statements about major cycle and minor cycle are correct? *1 point*

a. The major cycle represents the duration after which the full schedule pattern repeats
b. The minor cycle determines the smallest interval at which the scheduler wakes up and dispatches tasks
c. The major cycle must always be equal to the least common multiple (LCM) of all task periods
d. A task whose execution time is shorter than the minor cycle can be executed multiple times within the same minor cycle
e. The minor cycle duration must always be equal to the greatest common divisor (GCD) of all task execution times

- [x] a.
- [x] b.
- [x] c.
- [ ] d.
- [ ] e.

Yes, the answer is correct.
Score: 1
Accepted Answers:
a.
b.
c.

8) In a real-time system, consider a set of periodic tasks where each task has a phase and a response time. Which of the following statements are correct? *1 point*

a. The phase of a task specifies the time delay before the task's first release
b. Task response time is the duration between task arrival and completion
c. A periodic task with zero phase must start exactly at time t=0 in every scheduling cycle
d. A task having a larger phase has a reduced response time
e. Response time is the same as worst-case execution time for any task

- [x] a.
- [x] b.
- [ ] c.
- [ ] d.
- [ ] e.

Trending videos
Margo & Edith i...
ENG IN 89% 10:20 AM 15/04/2026

Real Time Operating System - ...
onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94
Ask Gemini
Google Gemini | ChatGPT | DeepSeek - Into the... | Generative AI for Be... | Home | Adobe | Course: Data Scienc... | HackNiche 4.0 | Dev... | All Bookmarks

8) In a real-time system, consider a set of periodic tasks where each task has a phase and a response time. Which of the following statements are correct? *1 point*

a. The phase of a task specifies the time delay before the task's first release
b. Task response time is the duration between task arrival and completion
c. A periodic task with zero phase must start exactly at time t=0 in every scheduling cycle
d. A task having a larger phase has a reduced response time
e. Response time is the same as worst-case execution time for any task

- [x] a.
- [x] b.
- [ ] c.
- [ ] d.
- [ ] e.

Partially Correct.
Score: 0.66
Accepted Answers:
a.
b.
c.

9) Which of the following statements correctly describes characteristics of event-driven and clock-driven (time-driven) real-time schedulers? *1 point*

a. Event-driven scheduling incurs lower runtime overhead than clock-driven scheduling
b. In event-driven scheduling, tasks are dispatched only at time-tick boundaries
c. Clock-driven scheduling decisions are made periodically at predefined instants of time
d. Event-driven schedulers typically use priorities to determine which task runs next
e. Clock-driven schedulers are usually preferred for highly periodic and safety-critical systems

- [ ] a.
- [ ] b.
- [x] c.
- [x] d.
- [x] e.

Yes, the answer is correct.
Score: 1
Accepted Answers:

Trending videos
Margo & Edith i...
ENG IN 89% 10:20 AM 15/04/2026

Real Time Operating System - ...
Ask Gemini
onlinecourses.nptel.ac.in/noc26_cs47/unit?unit=17&assessment=94
Google Gemini ChatGPT DeepSeek - Into the... Generative AI for Be... Home | Adobe Course: Data Scienc... HackNiche 4.0 | Dev... All Bookmarks

d.
e.

Yes, the answer is correct.
Score: 1

Accepted Answers:
c.
d.
e.

10) Suppose a real-time system deploys a foreground-background scheduler. An aperiodic task requiring 1000sec execution time becomes ready at time 0. There are three foreground period tasks as shown in the following table. What will be the completion time of the aperiodic task.

1 point

Task	Phase	Execution Time (in millisec)	Period (in millisec)	Deadline (in millisec)
T1	0	10	100	100
T2	0	20	150	150
T3	0	5	50	50

a. 1500 millisecs
b. 1800 millisecs
c. 2500 millisecs
d. 3000 millisecs
e. 3600 millisecs

a.
b.
c.
d.
e.

Yes, the answer is correct.
Score: 1

Accepted Answers:
a.

Trending videos
Margo & Edith i...
ENG IN 89% 10:20 AM 15/04/2026